IT 105 – Information Management I

Phase 4: Cloud Backup and Integration of Emerging Database Technologies

Platform Used

The database was deployed using Railway.app, a cloud-based platform that provides MySQL database hosting.

Cloud Deployment Details

Host: mysql.railway.internal

Port: 36521

Database Name: student_management_systemdb

Username: root

Deployment Procedure

1. Created a MySQL instance on Railway
2. Retrieved the database credentials (host, port, username)
3. Connected to the cloud database using MySQL Workbench
4. Imported the backup.sql file into the cloud database
5. Verified the database by executing SQL queries

Testing and Validation

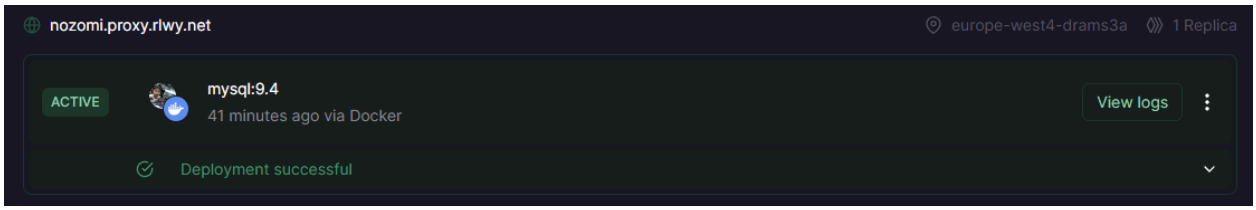
To verify the cloud deployment, a multi-table aggregate query was executed to confirm that all data was successfully imported and is accessible.

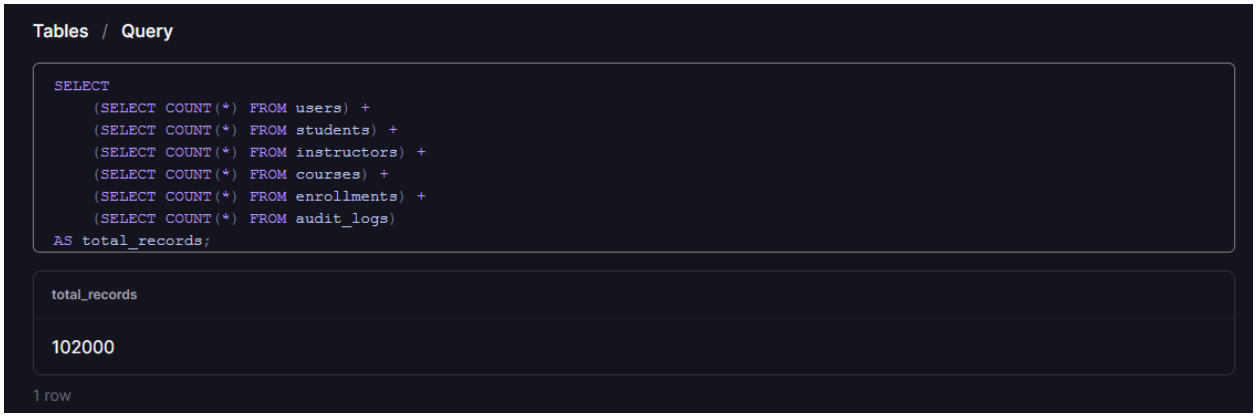
```
SELECT
   (SELECT COUNT(*) FROM users) +
   (SELECT COUNT(*) FROM students) +
   (SELECT COUNT(*) FROM instructors) +
   (SELECT COUNT(*) FROM courses) +
   (SELECT COUNT(*) FROM enrollments) +
   (SELECT COUNT(*) FROM audit_logs)
AS total_records;
```

Result:

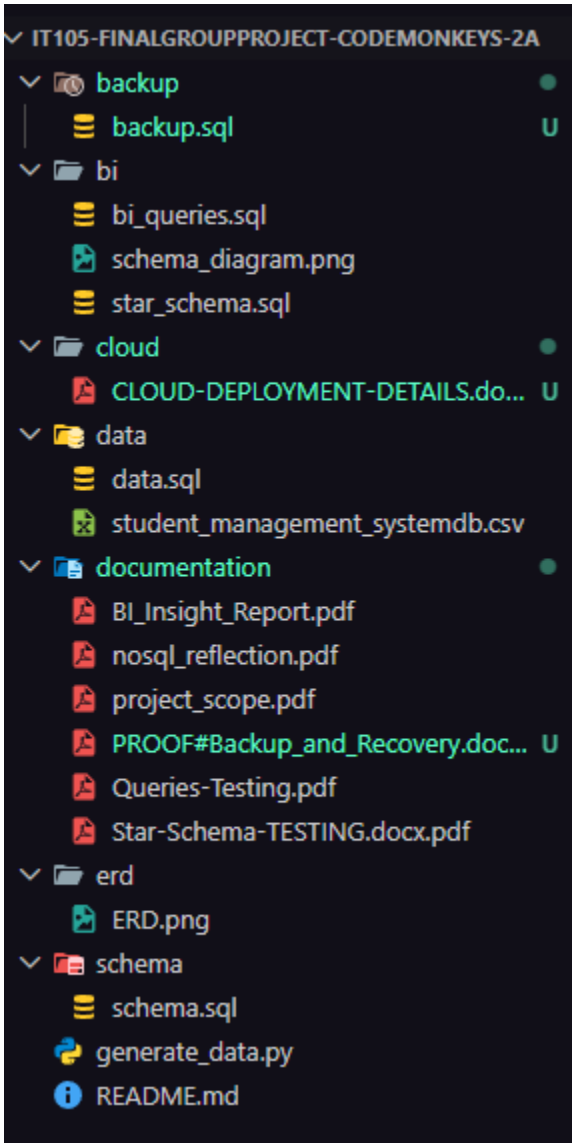
The query returned a total of 102,000 records, confirming that:

- The database is fully operational
- All tables were successfully imported
- Data integrity is maintained in the cloud environment





Project Structure



Conclusion

The cloud deployment was successfully completed and tested. The system is fully functional, accessible remotely, and correctly stores all project data.